Name: Priyanshu Digari
ROll no. : 89

Assignment-4
Information Theory and Coding

Q1. A code has minimum distance d = 4. Can it correct 2 errors?
Ans. Error correction capability:
t = (d - 1) / 2
Substitute d = 4:
t = (4 - 1) / 2 = (3 / 2) = 1
# Conclusion:
It can correct only 1 error
Therefore, it cannot correct 2 errors

Q2. Code A (d=3) vs Code B (d=5): Which is better?
Ans. Error correction capability:
- Code A: t = (3 - 1) / 2 = 1
- Code B: t = (5 - 1) / 2 =

Conclusion:
- Code B is better
- It can correct more errors (2 vs 1)
- Higher reliability and noise tolerance

Q3. Claim: "If a code detects 3 errors, it can also correct 3 errors."
Ans. Error detection condition:
- To detect s errors: $d \geq s + 1$
- If it detects 3 errors: $d \geq 4$
- Correction capability:
- $t = (d - 1) / 2 = 1$

Conclusion:
- It can detect 3 errors
- But can correct only 1 error
- Statement is false

Q4. (7,4) Hamming Code: Syndrome = 000. Is it always error-free?
Ans. Normal case:
Syndrome = 000 means no single-bit error detected
Edge

case:
Hamming code detects only single-bit errors
Two-bit errors may produce syndrome = 000
#Conclusion:
Not always error-free
Multiple-bit errors may go undetected

Q5. A code detects up to 2 errors. What is its minimum distance?
Ans. Detection condition:
$d \geq s + 1$
Given $s = 2$:
$d \geq 3$
#Conclusion:
Minimum distance = 3